

INTERNAL ENGINEERING PRACTICE SUMMARY

ENG-STD-SUMM-119 | Internal Engineering Summary - Hot Service Pump Requirements

Document No.	ENG-STD-SUMM-119	Issue	Issue 2, 2026-04-01
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SCOPE AND STATUS OF THIS DOCUMENT

This internal summary consolidates MRPL engineering practice for centrifugal pumps in hydrocarbon service above 250 degC. It is an internally authored interpretation prepared for training use and does not reproduce the text of any external standard.

This is an internal interpretation for training and prototype use. Where it differs from a governing external standard or statutory requirement, the external requirement prevails.

CONSOLIDATED REQUIREMENTS

Clause	Subject	Requirement
4.1	Seal selection	Pumps handling flammable liquid above 260 degC, or above the fluid auto-ignition temperature, shall use a dual seal arrangement. Arrangement 3 (dual pressurised) is the default selection; Arrangement 2 (dual unpressurised) requires documented justification.
4.2	Seal support system	Arrangement 3 seals shall be supported by a Plan 53B or Plan 53C barrier fluid system with level and pressure transmitted to the DCS. Manually replenished Plan 52 or Plan 53A reservoirs shall not be used as the sole protection on services above 260 degC.
4.3	Alignment	Shaft alignment shall be verified by laser method. Maximum permissible parallel offset is 0.05 mm and maximum angular deviation is 0.05 mm per 100 mm of coupling span, measured at operating temperature where hot alignment is specified.
5.1	Condition monitoring	Overall vibration velocity shall be trended monthly. A reading at or above 7.1 mm/s RMS constitutes an alert condition and shall generate a corrective work order within 24 hours. A reading at or above 11.0 mm/s RMS requires the pump to be removed from service at the earliest safe opportunity.
5.2	Bearing temperature	Bearing metal or oil temperature shall alarm at 85 degC and trip at 95 degC for pumps of 150 kW and above.
6.1	Spares holding	Criticality A rotating equipment shall hold a minimum of two complete seal kits per pump pair on site. Stock below minimum shall be reported in the monthly reliability review.
6.2	Standby availability	Where a pump pair serves a continuous process duty, the standby unit shall be maintained warm, lined up and on auto-start at all times, and shall be test run for not less than 30 minutes every 14 days.

APPLICATION TO CDU-2 COLUMN BOTTOMS PUMPS

Clause	Requirement	P-2104A/B as installed	Compliant?
4.1	Dual seal, Arrangement 3 default	Category 1, Arrangement 1 (single unpressurised)	NO
4.2	Plan 53B or 53C with DCS transmission	Plan 52 external reservoir, manually replenished, no level switch	NO
4.3	Laser alignment, 0.05 mm parallel max	0.03 mm as left on 2026-09-02; was 0.19 mm as found	YES (as left)
5.1	Corrective work order within 24 h of a 7.1 mm/s reading	7.8 mm/s recorded 2026-07-31, no work order raised	NO
5.2	Bearing alarm 85 degC, trip 95 degC	Alarm configured at 85 degC; no trip configured	PARTIAL
6.1	Two seal kits per pump pair in stock	1 x SK-682-2104 and 0 x SK-682-2104-A3 in stock	NO

Clause	Requirement	P-2104A/B as installed	Compliant?
6.2	Standby warm, lined up, test run every 14 days	Complied; last test run of standby 2026-08-18	YES